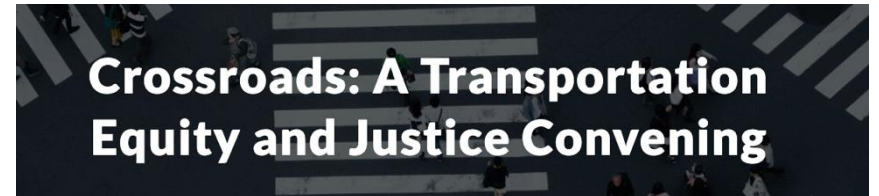


Beyond Accessibility Disparity: Understanding Transport Poverty in the U.S. through Lived Experiences

Md Hamidur Rahman, Alex Karner

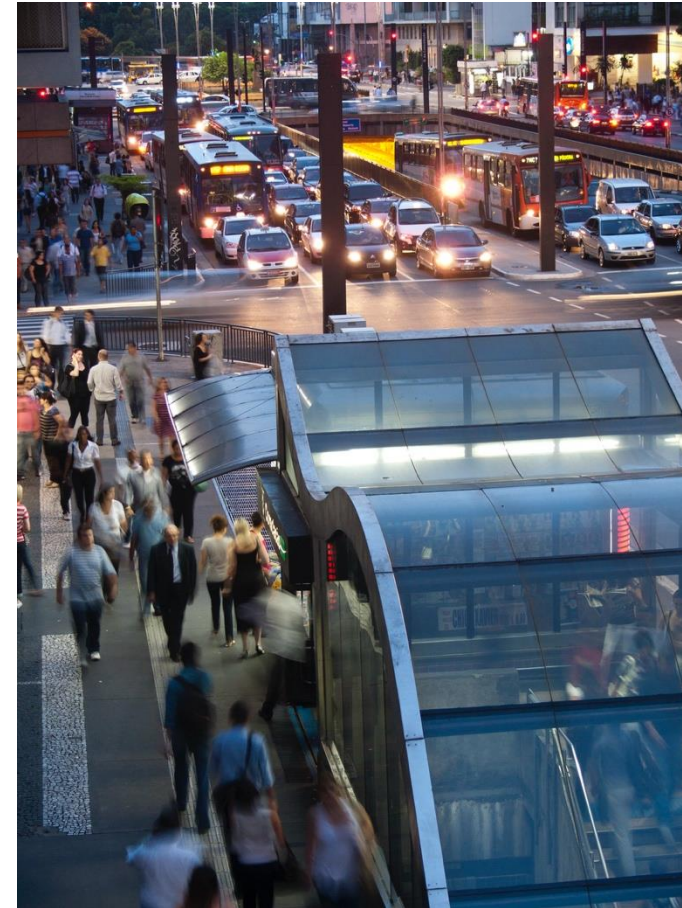
Graduate Program in Community and Regional Planning



January 15, 2026
Martin Luther King Jr. Memorial Library, Washington D.C.

Why this Study?

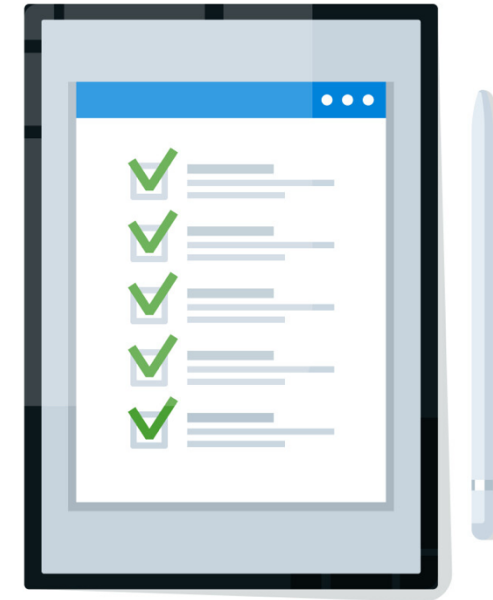
- ❑ Traditional travel surveys fail to capture people's unmet travel needs.
- ❑ Prior studies have focused on: specific disadvantaged groups, specific modes and places, specific transportation problems.
- ❑ Growing literature on measuring travel or mobility problems : Lättman et al. (2019), Murphy et al. (2021), Singer and Martens (2023), De Vos (2024), Ettema et al. (2025), Palm et al. (2025).



1 in 4 Americans are unable to regularly and reliably access transportation they require to meet their daily needs (FTA, 2023).

Objective

- ✓ Create a survey instrument that can be used to measure perceived travel experiences, perception of access, activity participation, and life outcomes across a population.



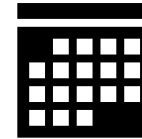
Data Collection: Pilot Study



Online Survey



Qualtrics Panel on
US General Population



Jul 15-Aug 15 (2025)



Total 670 Sample



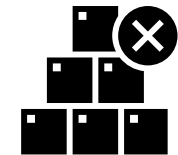
Quotas:
Gender, Age, Race,
Ethnicity and Region



54 Questions



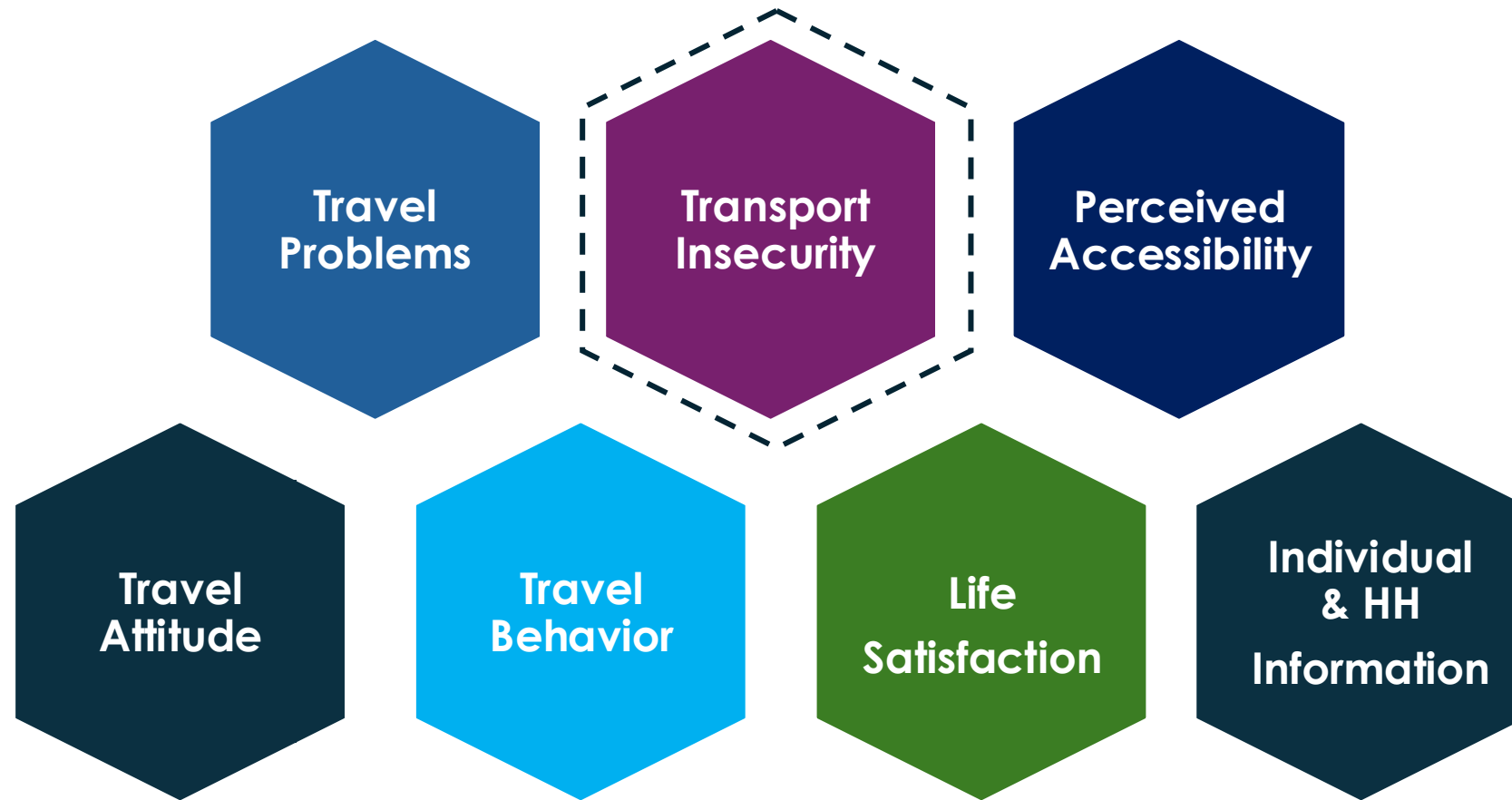
completion time
~ 12 mins



2 attention check,
completion time,
straight-lining

***616** Final Sample

Survey Data Collection: 7 Key Themes



Transport Insecurity



USDOT defines Transportation insecurity

“when people are unable to get to where they need to go to meet the needs of their daily life regularly, reliably, and safely”.

- Inspired from the measure of "Food Security" of U.S. Department of Agriculture (USDA), "Housing Insecurity" of Department of Housing and Urban Development (HUD).
- Recent study *National Academies of Sciences, Engineering, and Medicine (2025)* recommended pilot testing of [Transportation Insecurity measure](#).

Results

Transport Insecurity Scale

- We used validated 6-items measurement scale of transport insecurity developed by Murphy et al. (2024).

To get to the places they need to go, people might walk, bike, take a bus, train or taxi, drive a car, or get a ride.

1. In the past 30 days, how often did you have to **reschedule an appointment** because of a problem with transportation?

2. In the past 30 days, how often did you **skip going somewhere** because of a problem with transportation?

3. In the past 30 days, how often were you **not able to leave the house** when you wanted to because of a problem with transportation?

4. In the past 30 days, how often did you **feel bad** because you did not have the transportation you needed?

5. In the past 30 days, how often did you **worry about inconveniencing your friends, family, or neighbors** because you needed help with transportation?

6. In the past 30 days, how often did problems with transportation affect your **relationships with others**?

Response Category: Often – Sometimes – Never

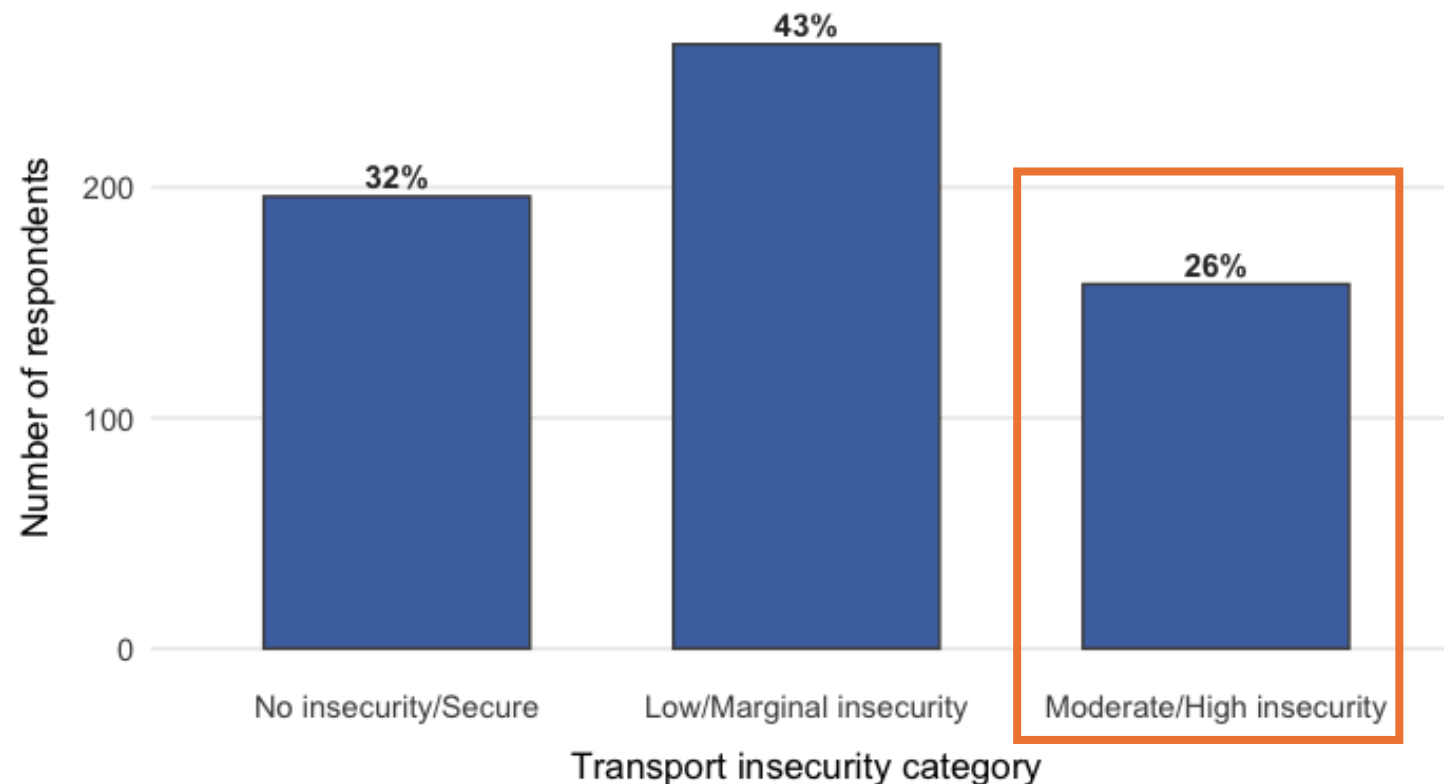
Transport Security Index (TSI) Score

TSI Scoring:

- 0= Never, 1= Sometimes, 2= Often
- Sum of scores
 0-1 (No insecurity/Secure),
 2-6 (Low/Marginal Insecurity)
 7-12 (Moderate/High Insecurity)

Distribution of Transport Insecurity Categories

Share of respondents by insecurity level



Proportion of adults over 18 experiencing different level of transport insecurity in 2025.

Transport Security Index (TSI) for population segments

Population Segment	TSI Score		Significance
	Mean	SD	(t-test)
Men	5.122	3.838	ns
Women	4.559	3.974	ns
Non-White	5.250	3.805	p<0.01
White	4.394	3.988	p<0.01
Employed	5.503	3.849	p<0.001
Not employed	3.867	3.811	p<0.001
With disability	5.994	3.678	p<0.001
Without disability	4.408	3.916	p<0.001
High income (≥\$120k)	6.170	4.103	p<0.001
Middle income (\$40–120k)	4.857	3.853	p<0.001
Low income (<\$40k)	4.137	3.773	p<0.001
Multimodal traveler	6.340	3.186	p<0.001
Car-less	5.514	3.837	p<0.001
Daily driver	3.854	3.981	p<0.001
No car access	5.546	3.715	p<0.05
Has car access	4.682	3.943	p<0.05
Urban	5.492	4.005	p<0.01
Suburban	4.325	3.892	p<0.01
Rural	4.389	3.520	p<0.01

- Average TSI is significantly higher among **non-White respondents, people with disabilities, multimodal travelers, and those living in urban areas.**
- Surprisingly, employed and high-income respondents report significantly higher TSI than unemployed and low-income groups.
- No statistically significant difference is observed in terms of gender.

Consequences of Transport Insecurity

Respondent with **Moderate/High Transport Insecurity** (N=156) level:

- Have lower # unique of activities in past 7 days (average 3.42 activities)
- Highest unmet activity needs (81.9% of of 156 respondents)
- Lowest average perceived accessibility (3.34 on scale of 5)
- Lowest average life satisfaction (5.85 one a scale of 10)

Key Findings and Next Steps

- A low percentage of the sample experienced **medium-high transport insecurity (26%)**.
- Population groups like **non-White, with limited abilities, multimodal travelers, urban residents** experienced transport insecurity frequently.
- Higher transport insecurity results in higher unmet activity needs, low perceived access to opportunities, and lower life satisfaction of respondents.
- **Next Steps:**
 - Testing the **validity and reliability of the survey items** on a national and local scale
 - Policy relevancy check i.e., incorporating validated metrics into existing performance measurement.

Thank You!



Md Hamidur Rahman

PhD Student in Community and Regional Planning

hamidur.rahman@utexas.edu

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Extra Slides for Q/A

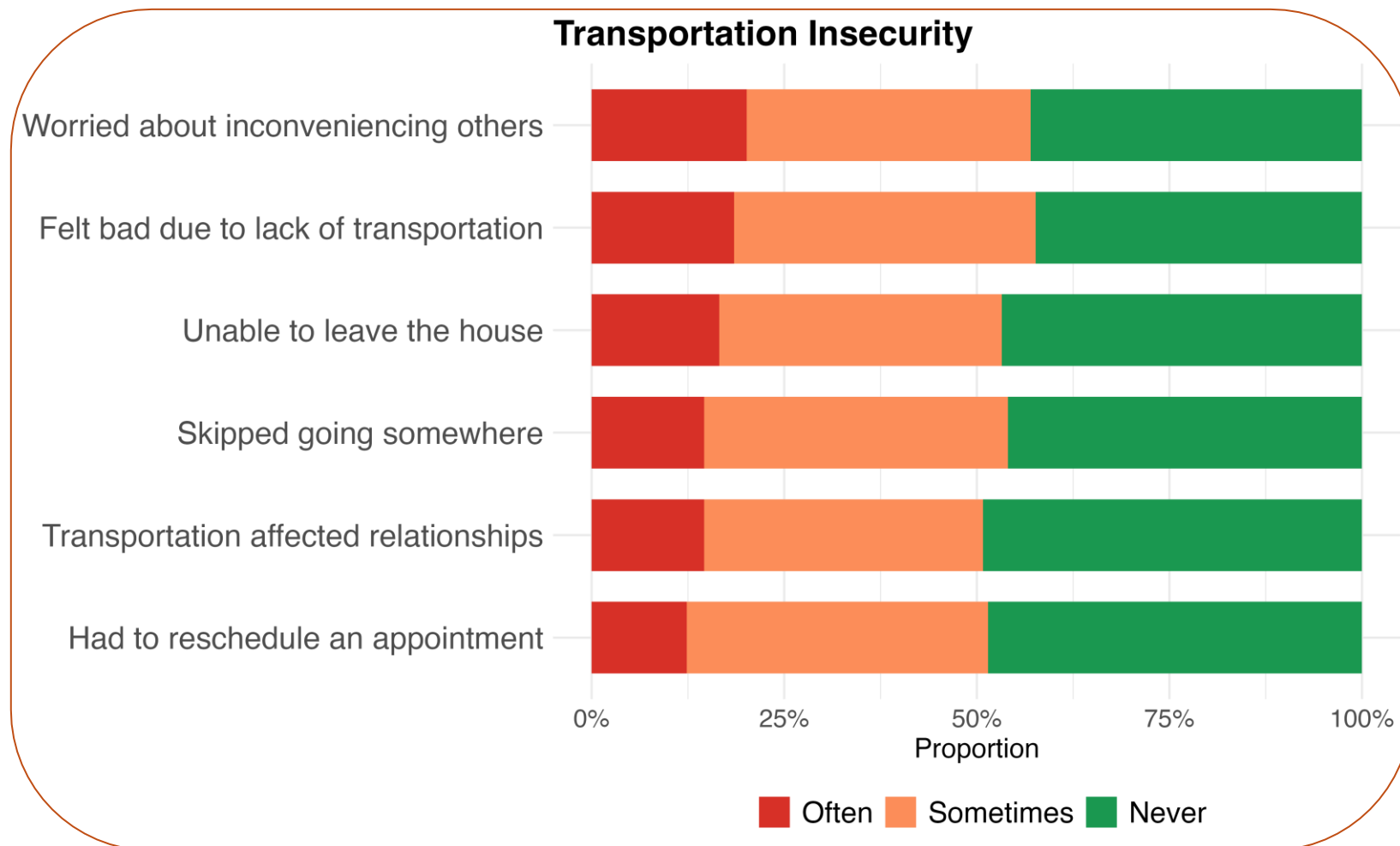
Surevey [Link](#)

https://drive.google.com/file/d/1_OYQCdVc9BoeuiSwNY4iywrxroCZVSAo/view?usp=sharing

Transport Insecurity Scale

Survey response showed:

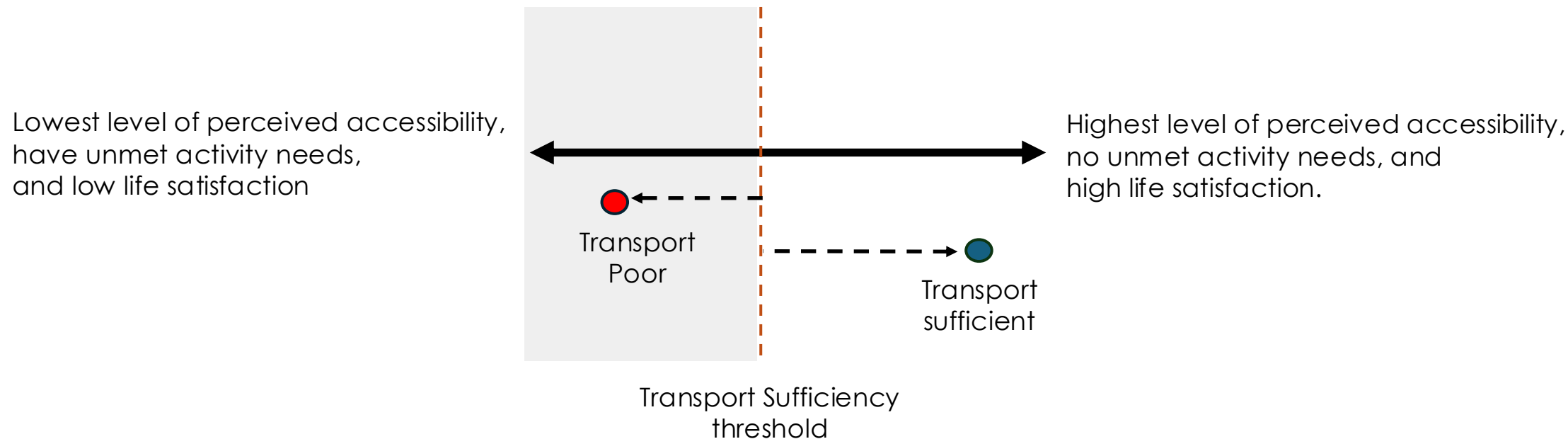
- Substantial share of respondents report experiencing transportation insecurity at least sometimes,
- Worrying, feeling bad, and unable to leave house when wanted emerging as the most common challenges.



Consequences of Transport Insecurity

	No insecurity/ Secure (N=214)	Low/Marginal Insecurity (N=246)	Moderate/ High Insecurity (N=156)	Sig. (ANOVA)
# Unique of activities in past 7 days (average)	3.52	3.53	3.42	0.924
Unmet Activity Need (% of Yes)	16.1	68.0	81.9	<0.001
Perceived adequacy of opportunity accessibility (average of scale 1-5)	4.30	3.72	3.34	<0.001
Satisfaction with Life as whole (average on Scale 1-10)	6.71	5.89	5.85	<0.001

Towards A Framework of Transport Sufficiency

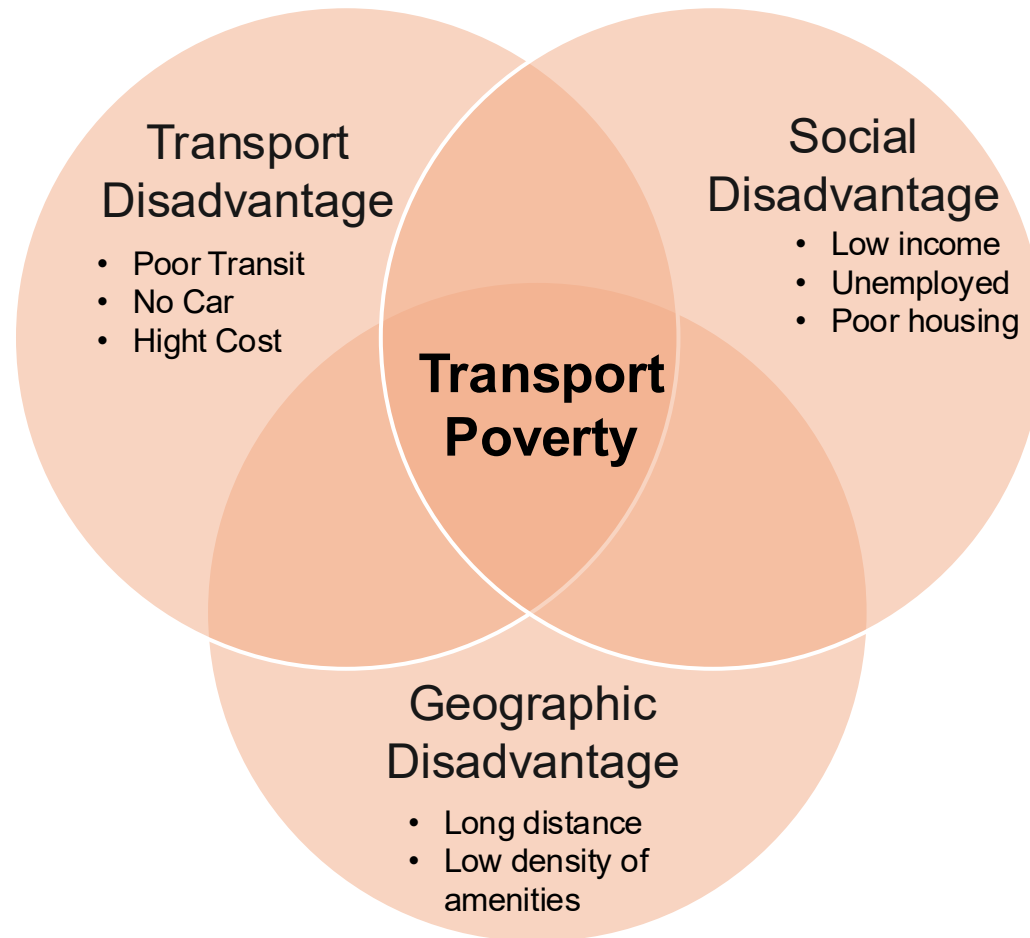


Measuring Transport Insecurity

Thinking about the past 30 days:

	Never	Sometimes	Often
How often did you have to reschedule an appointment because of a problem with transportation?	☐	☐	☐
How often did you skip going somewhere because of a problem with transportation?	☐	☐	☐
How often were you unable to leave the house when you wanted to because of a problem with transportation?	☐	☐	☐
How often did you feel bad because you did not have the transportation you needed?	☐	☐	☐
How often did you worry about inconveniencing your friends, family, or neighbors because you needed help with transportation?	☐	☐	☐
How often did problems with transportation affect your relationships with others?	☐	☐	☐

Transport Poverty

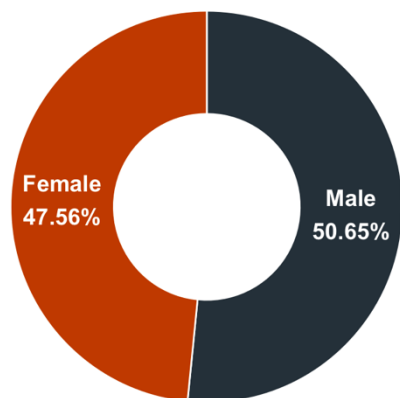


Transport Poverty

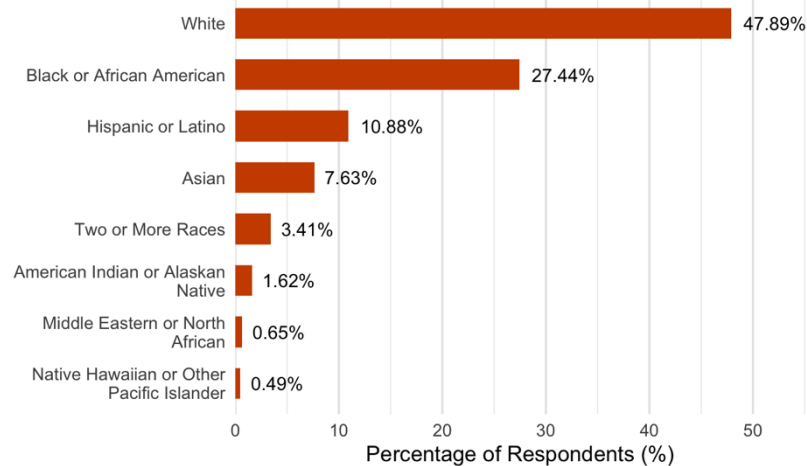
An individual experiences transport poverty if he or she lacks access to **adequate means of transport**, limiting a person's potential mobility to reach or participation in activities (Jeekel and Martens, 2017).

Sample Distribution

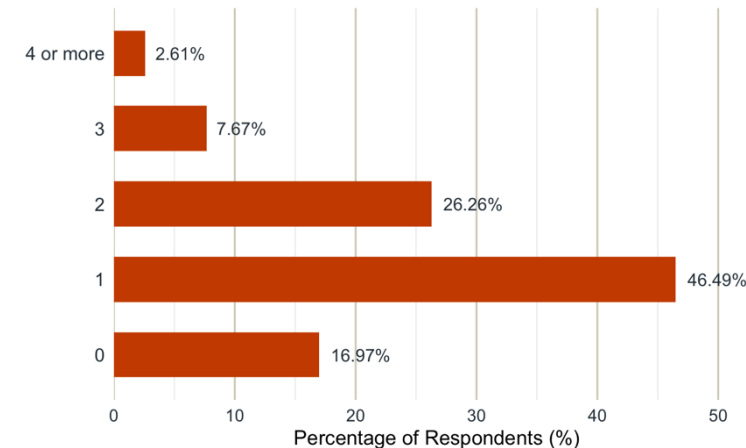
Gender Distribution of Respondents



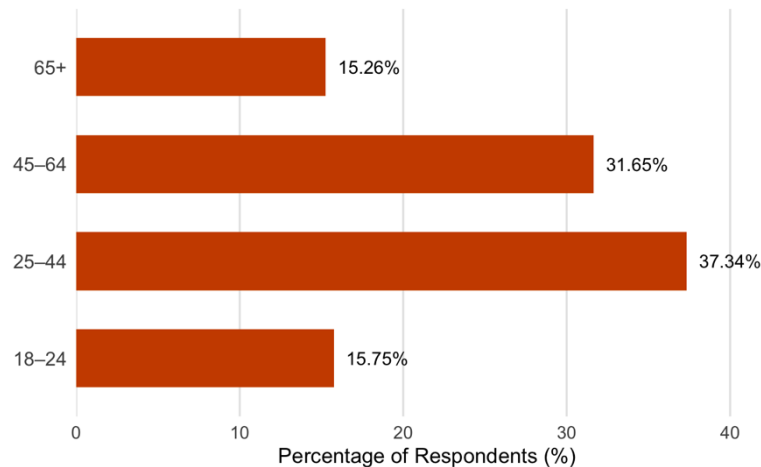
Race/Ethnicity Distribution of Respondents



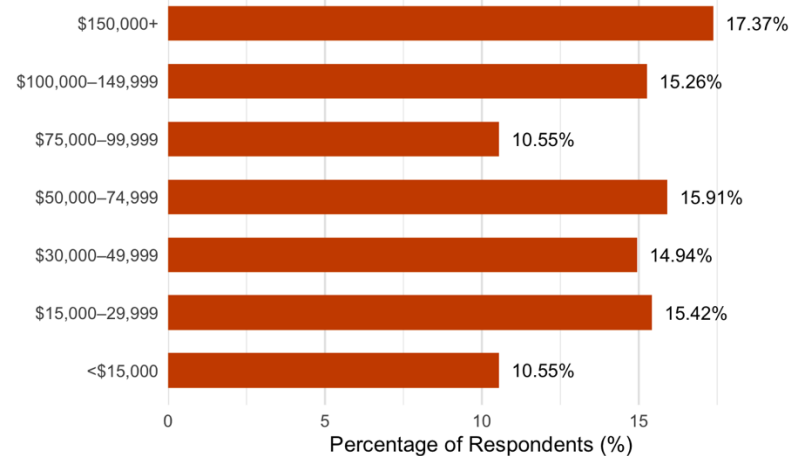
Household Vehicle Ownership



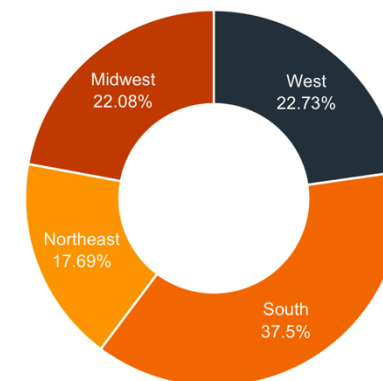
Age Distribution of Respondents



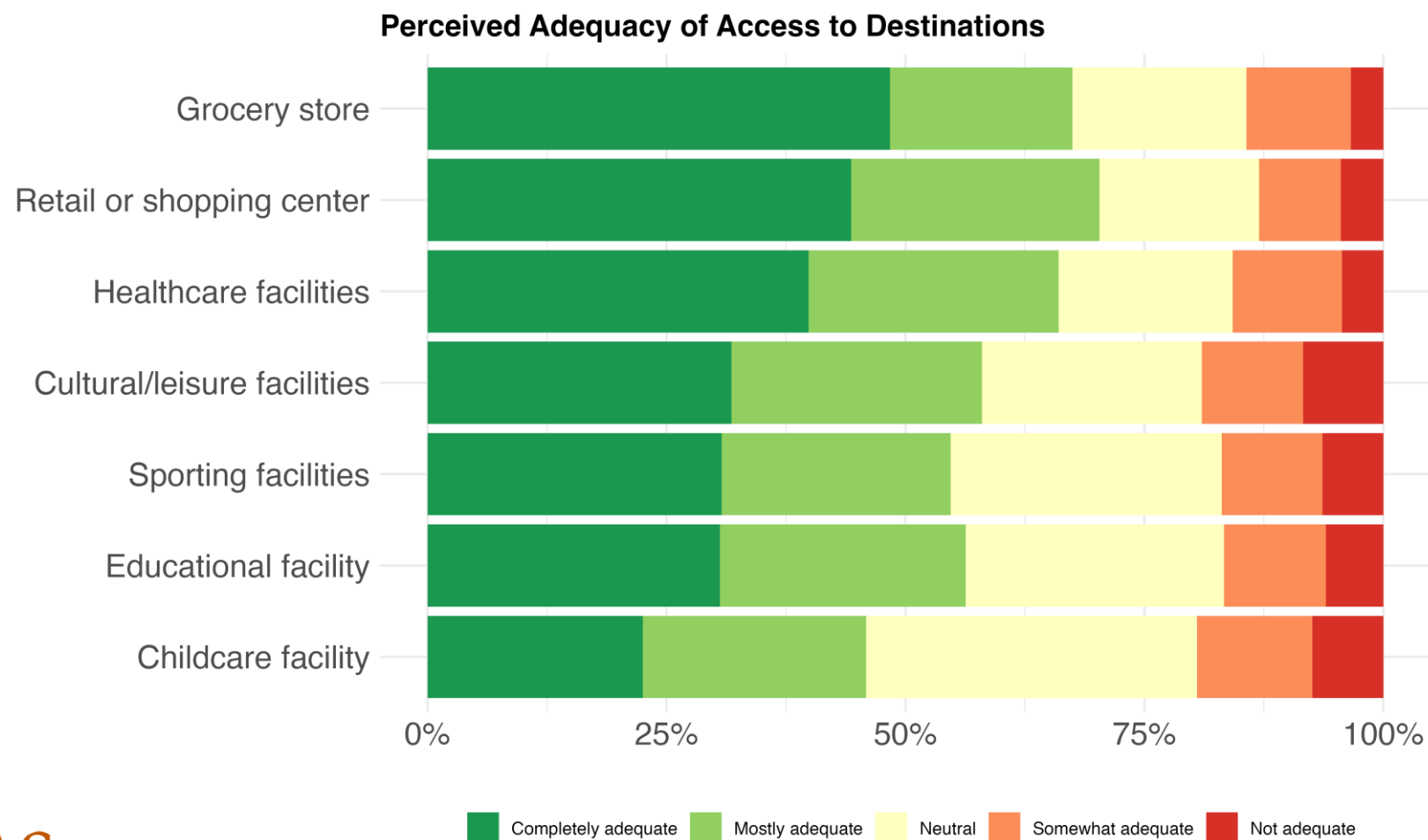
Household Income Distribution



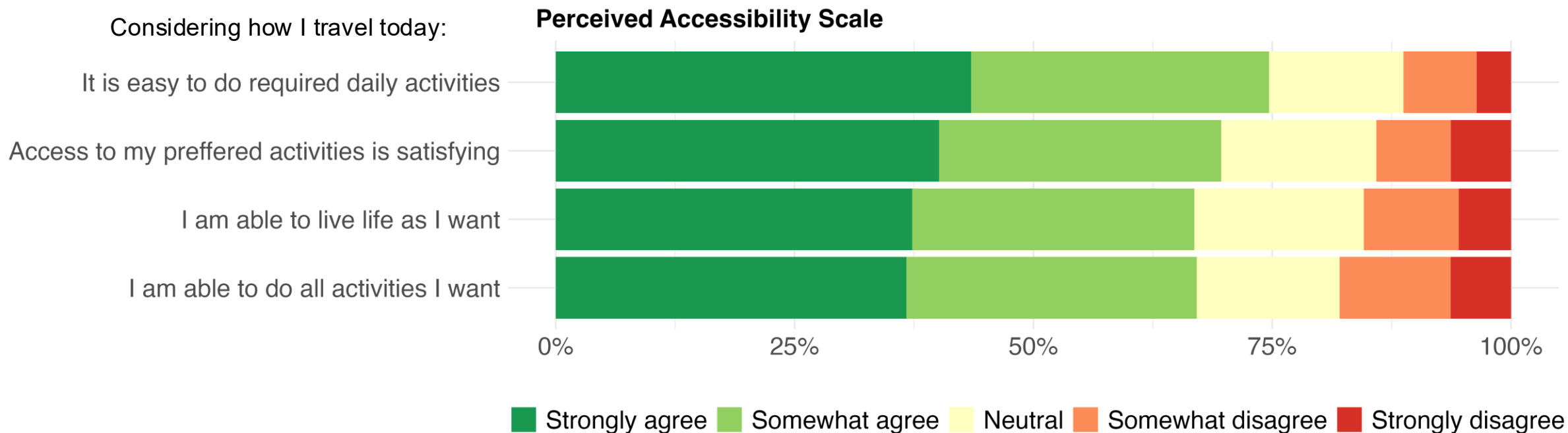
Region of Residence



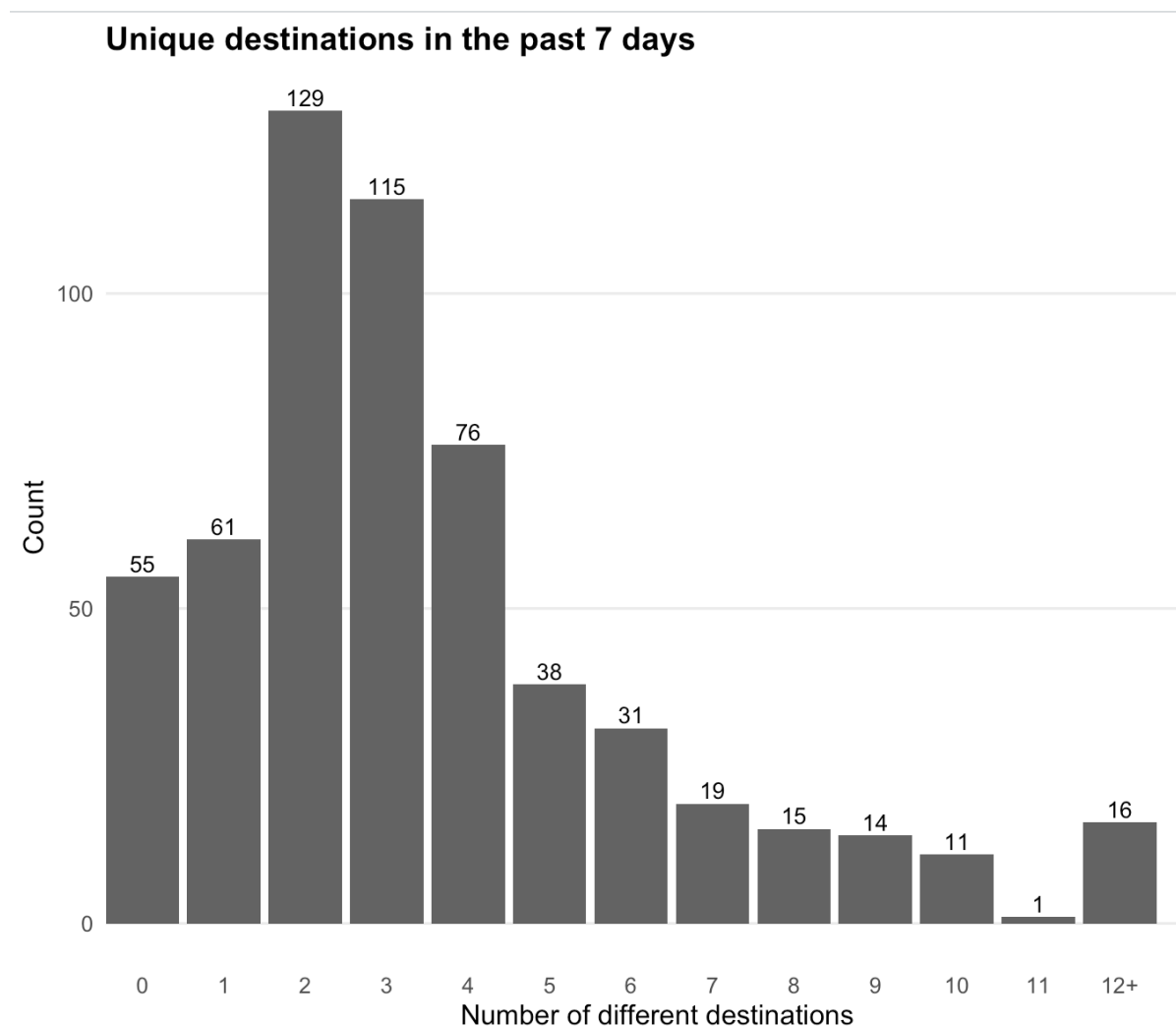
Access to destinations



Overall perception of access



Descriptive Statistics



Descriptive Statistics

